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Division- SY1 ENTC

1. Perform following String operations: a. substring b. palindrome c. compare d. copy e. reverse.

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#include <stdio.h>

int getLen(char str[]){
    int len = 0;
    while(str[len] != '\0'){
        len++;
    }
    return len;
}

void substring(char str[], int pos, int len){
    char sub[100];
    int i;
    for(i=0; i<len && str[pos+i] != '\0'; i++){
        sub[i] = str[pos+i];
    }
    sub[i] = '\0';
    printf("Substring: %s\n", sub);
}

void checkPalindrome(char *str){
    char *p1 = str;
    char *p2 = str;

    while(*p2 != '\0'){
        p2++;
    }
    p2--;

    int flag = 1;
    while(p1 < p2){
        if(*p1 != *p2){
            flag = 0;
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            break;
        }
        p1++;
        p2--;
    }

    if(flag == 1)
        printf("String is Palindrome\n");
    else
        printf("String is NOT Palindrome\n");
}

void compareStrings(char s1[], char s2[]){
    int i = 0;
    int flag = 1;

    while(s1[i] != '\0' || s2[i] != '\0'){
        if(s1[i] != s2[i]){
            flag = 0;
            break;
        }
        i++;
    }

    if(flag == 1)
        printf("Strings are Matching\n");
    else
        printf("Strings are Not Matching\n");
}

void copyString(char *src, char *dest){
    while(*src != '\0'){
        *dest = *src;
        src++;
        dest++;
    }
    *dest = '\0';
}

void reverseString(char str[]){
    int len = getLen(str);
    char rev[100];
    int i;
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    for(i = 0; i < len; i++){
        rev[i] = str[len - 1 - i];
    }
    rev[len] = '\0';
    printf("Reversed string: %s\n", rev);
}

int main(){
    char str1[100], str2[100], copiedStr[100];
    int choice, pos, len;

    printf("Enter string: ");
    scanf("%s", str1);

    while(1){
        printf("\n--- MENU ---\n");
        printf("1. Substring\n");
        printf("2. Palindrome check\n");
        printf("3. Compare\n");
        printf("4. Copy\n");
        printf("5. Reverse\n");
        printf("6. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        if(choice == 1){
            printf("Enter starting index: ");
            scanf("%d", &pos);
            printf("Enter length: ");
            scanf("%d", &len);
            substring(str1, pos, len);
        }
        else if(choice == 2){
            checkPalindrome(str1);
        }
        else if(choice == 3){
            printf("Enter second string: ");
            scanf("%s", str2);
            compareStrings(str1, str2);
        }
        else if(choice == 4){
            copyString(str1, copiedStr);
            printf("Copied string: %s\n", copiedStr);
        }
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        else if(choice == 5){
            reverseString(str1);
        }
        else if(choice == 6){
            break;
        }
        else{
            printf("Invalid choice!\n");
        }
    }

    return 0;
}
```